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Assignment 2
Due on Feb 25, 2020

One researcher found one organism has a very interesting property. After many steps' purification, he was able to get a small amount of protein. Then he used Mass spectrometry (MS) to find out the sequence and the molecular weight of the protein. The protein sequence is provided in the form of single-letter amino acid code.

1 medaknikkg papfypledg tageqlhkam kryalvpgti aftdahievd ityaeyfems
61 vrlaeamkry glntnhrivv csenslqffm pvlgalfigv avapandiyn erellnsmgi
121 sqptvvfvsk kglqkilnvq kklpiiqkii imdsktdyqg fqsmytfvts hlppgfneyd
181 fvpesfdrdk tialimnssg stglpkgval phrtacvrfs hardpifgnq iipdtailsv
241 vpfhhgfgmf ttlgylicgf rvvlmyrfee elflrslqdy kiqsallvpt lfsffakstl
301 idkydlsnlh eiasggapls kevgeavakr fhlpgirqgy gltettsail itpegddkpg
361 avgkvvpffe akvvdldtgk tlgvnqrgel cvrgpmimsg yvnnpeatna lidkdgwlhs
421 gdiaywdede hffivdrlks likykgyqva paelesillq hpnifdagva glpdddagel
481 paavvvlehg ktmtekeivd yvasqvttak klrggvvfvd evpkgltgkl darkireili
541 kakkggkiav

He needs to make more protein to further characterize it. Unfortunately, he does not have enough organisms to provide him enough protein for the characterization.

You need to help him by doing the following things:

(a) Propose a method to make more protein (as detailed as possible)

DNA sequence can be obtained by amino acid code.
Use DNA Synthesizers to create specific DNA molecules
Polymerase Chain Reaction (PCR) to obtain more DNA fragment.
1. Initialization step: This step consists of heating the reaction to a temperature of 94-96°C , which is held for 19 minutes. It is only required for DNA polymerases that require heat activation by hot-start PCR.
2.Denaturation step: This step is the first regular cycling event and consists of heating the reaction to 94 98°C for 20-30 seconds. It causes DNA melting of the DNA template by disrupting the hydrogen bonds between complementary bases, yielding single-stranded DNA molecules.
3.Annealing step: The reaction temperature is lowered to 50-65°C for 20-40 seconds allowing annealing of the primers to the single-stranded DNA template. Typically the annealing temperature is about 3-5 degrees Celsius below the Tm of the primers used. Stable DNA-DNA hydrogen bonds are only formed when the primer sequence very closely matches the template sequence. The polymerase binds to the primer-template hybrid and begins DNA synthesis.

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4.Extension/elongation step: The temperature at this step depends on the DNA polymerase used; Taq polymerase has its optimum activity temperature at 75-80°C, and commonly a temperature of 72°C is used with this enzyme. At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTPs that are complementary to the template in 5' to 3' direction, condensing the 5'-phosphate group of the dNTPs with the 3'-hydroxyl group at the end of the nascent (extending) DNA strand. The extension time depends both on the DNA polymerase used and on the length of the DNA fragment to be amplified. As a rule-of-thumb, at its optimum temperature, the DNA polymerase will polymerize a thousand bases per minute.

5.Final elongation: This single step is occasionally performed at a temperature of 70-74°C for 5-15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully extended.

6.Final hold: This step at 4-15°C for an indefinite time may be employed for short-term storage of the reaction.

This cycle repeats 25-35 times in a typical PCR reaction. After the last cycle, the reaction was maintained at 72℃ for 10-30 minutes. The primers are extended completely and the single chain products are annealed into double chains

6. ~~Cut-open~~ plasmid (eg: pUC19) with restriction ~~enzymes, leaves~~ "Sticky ends". what plasmid

7. Insert the Plasmid with Recombinant DNA into a new bacterium.

~~8. Screen the bacteria~~ reproduces itself and the plasmid. All descendents express the inserted genes

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9. Protein purification

(b) Tell him the molecular weight of this protein

The molecular weight of this protein = The total MW of all amino acid – 549*18(The MW of H_2O)

The protein weighs 60.65 00kilodaltons (https://www.bioinformatics.org/sms/prot_mw.html)

(c) Convert the protein sequence to corresponding DNA sequence (Please specific which codon you are using)

5' → 3' direction(coding strand):

atggaagatgcgaaaaacattaaaaaaggcccggcgccgttttatccgctggaagatggc
accgcgggcgaacagctgcataaagcgatgaaacgctatgcgctggtgccgggcaccatt
gcgtttaccgatgcgcatattgaagtggatattacctatgcggaatattttgaaatgagc
gtgcgcctggcggaagcgatgaaacgctatggcctgaacaccaaccatcgcattgtggtg
tgcagcgaaaacagcctgcagtttttatgccggtgctgggcgcgctgtttattggcgtg
gcggtggcgccggcgaacgatatttataacgaacgcgaactgctgaacagcatgggcatt
agccagccgaccgtggtgtttgtgagcaaaaaaggcctgcagaaaattctgaacgtgcag
aaaaaactgccgattattcagaaaattattattatggatagcaaaaccgattatcagggc
tttcagagcatgtatacctttgtgaccagccatctgccgccgggctttaacgaatatgat
tttgtgccggaaagctttgatcgcgataaaaccattgcgctgattatgaacagcagcggc

agcaccggcctgccgaaaggcgtggcgctgccgcatcgcaccgcgtgcgtgcgctttagc
catgcgcgcgatccgattttggcaaccagattattccggataccgcgattctgagcgtg
gtgccgtttcatcatggctttggcatgtttaccaccctgggctatctgatttgcggcttt
cgcgtggtgctgatgtatcgctttgaagaagaactgtttctgcgcagcctgcaggattat
aaaattcagagcgcgctgctggtgccgaccctgtttagcttttttgcgaaaagcaccctg
attgataaatatgatctgagcaacctgcatgaaattgcgagcggcggcgcgccgctgagc
aaagaagtgggcgaagcggtggcgaaacgctttcatctgccgggcattcgccagggctat
ggcctgaccgaaaccaccagcgcgattctgattaccccggaaggcgatgataaaccgggc
gcggtgggcaaagtggtgccgttttttgaagcgaaagtggtggatctggataccggcaaa
accctgggcgtgaaccagcgcggcgaactgtgcgtgcgcggcccgatgattatgagcggc
tatgtgaacaacccggaagcgaccaacgcgctgattgataaagatggctggctgcatagc
ggcgatattgcgtattgggatgaagatgaacatttttttattgtggatcgcctgaaaagc
ctgattaaatataaaggctatcaggtggcgccggcggaactggaaagcattctgctgcag
catccgaacattttgatgcgggcgtggcgggcctgccggatgatgatgcgggcgaactg
ccggcggcggtggtggtgctggaacatggcaaaaccatgaccgaaaaagaaattgtggat
tatgtggcgagccaggtgaccaccgcgaaaaaactgcgcggcggcgtggtgtttgtggat
gaagtgccgaaaggcctgaccggcaaactggatgcgcgcaaaattcgcgaaattctgatt
aaagcgaaaaaaggcggcaaaattgcggtg(taa)*

* taa is a stop codon

Codon Table*

AmAcid	Codon	Number	/1000	Fraction	AmAcid	Codon	Number	/1000	Fraction
Gly	GGG	50527	11.12	0.15	Trp	TGG	65630	14.45	1
Gly	GGA	39036	8.59	0.12	End	TGA	4428	0.97	0.3
Gly	GGT	114185	25.14	0.34	Cys	TGT	23461	5.17	0.45
Gly	GGC	130043	28.63	0.39	Cys	TGC	28747	6.33	0.55
Glu	GAG	83804	18.45	0.32	End	TAG	1172	0.26	0.08
Glu	GAA	179460	39.51	0.68	End	TAA	9006	1.98	0.62
Asp	GAT	146794	32.32	0.63	Tyr	TAT	75774	16.68	0.58
Asp	GAC	87759	19.32	0.37	Tyr	TAC	55847	12.3	0.42
Val	GTG	115687	25.47	0.36	Leu	TTG	60322	13.28	0.13
Val	GTA	51020	11.23	0.16	Leu	TTA	62823	13.83	0.13
Val	GTT	86572	19.06	0.27	Phe	TTT	100128	22.05	0.57
Val	GTC	67356	14.83	0.21	Phe	TTC	74885	16.49	0.43
Ala	GCG	146264	32.2	0.34	Ser	TCG	39546	8.71	0.15
Ala	GCA	93390	20.56	0.22	Ser	TCA	35837	7.89	0.13
Ala	GCT	73677	16.22	0.17	Ser	TCT	42367	9.33	0.16
Ala	GCC	113412	24.97	0.27	Ser	TCC	40365	8.89	0.15

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Arg	AGG	7423	1.63	0.03	Arg	CGG	25751	5.67	0.1
Arg	AGA	12345	2.72	0.05	Arg	CGA	16607	3.66	0.07
Ser	AGT	41544	9.15	0.15	Arg	CGT	93997	20.7	0.37
Ser	AGC	70867	15.6	0.26	Arg	CGC	96053	21.15	0.38
Lys	AAG	51685	11.38	0.25	Gln	CAG	130898	28.82	0.66
Lys	AAA	156169	34.38	0.75	Gln	CAA	67129	14.78	0.34
Asn	AAT	84846	18.68	0.46	His	CAT	57585	12.68	0.57
Asn	AAC	98018	21.58	0.54	His	CAC	43743	9.63	0.43
Met	ATG	123604	27.21	1	Leu	CTG	231373	50.94	0.49
Ile	ATA	24233	5.34	0.09	Leu	CTA	18067	3.98	0.04
Ile	ATT	135873	29.92	0.5	Leu	CTT	51442	11.33	0.11
Ile	ATC	111878	24.63	0.41	Leu	CTC	48147	10.6	0.1
Thr	ACG	63696	14.02	0.26	Pro	CCG	101467	22.34	0.51
Thr	ACA	35995	7.93	0.15	Pro	CCA	38663	8.51	0.2
Thr	ACT	43256	9.52	0.18	Pro	CCT	32678	7.19	0.17
Thr	ACC	103121	22.7	0.42	Pro	CCC	24383	5.37	0.12

* Table was generated using all the E. coli coding sequences in GenBank.

(d) What is the putative function of this protein? (using protein blast tools in one of the provided bioinformatics databases)

Based on the results of the Blast tool, this protein is a luciferase (Sequence ID: AAA88784.1). It can uses luciferin as a substrate, oxidizing it to oxyluciferin in a reaction that utilizes molecular oxygen and ATP, and liberates light (Firefly luciferase - 560 nm (Wilson and Hastings, 1998; Fraga, 2008); Bacterial luciferase - 490 nm (Waidmann et al., 2011))

(e) Give at least three applications of this protein

1. If the cells are labeled with luciferase, we can use a modified optical microscopes to observe it in vivo without harming the animals.

2. It can be used for luciferase assay, which is used to determine if a protein can activate or repress the expression of a target gene.

3. Luciferase can also be used to detect the level of cellular ATP in cell viability assays or for kinase activity assays. Luciferase can act as an ATP sensor protein through biotinylation.